The Perils of Bilateral Sovereign Debt

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Official Sovereign Debt

- A large share of sovereign borrowing takes the form of official debt
  - … Multilaterals, development banks, other governments

- Emergence of new bilateral creditors **outside** the Paris Club IDS data
  - … with claims to seniority and sometimes **confidential** terms
  - … esp. when Central Banks are involved

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- How does the presence of a large senior lender affect sovereign debt markets?
- What are its welfare implications for borrowing governments?

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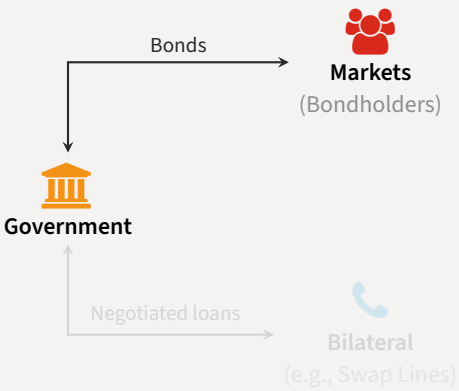
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Evaluating Senior Official Creditors

Quantitative sovereign debt model with

- Competitive creditors in private markets
- Large bilateral lender
  1. Superior enforcement [de-facto seniority]
  2. Bargained terms [price and quantity]
  3. Short-maturity loans
- Prime example: Central Bank swap lines (Horn et al., 2021)
- Focus on the interaction between both funding sources



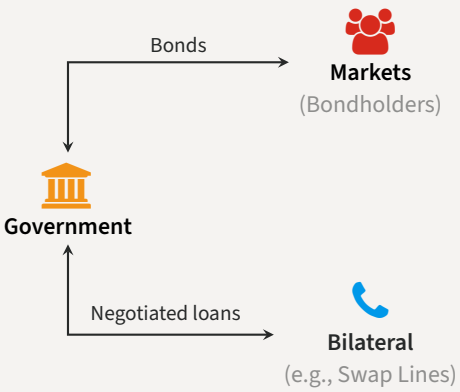
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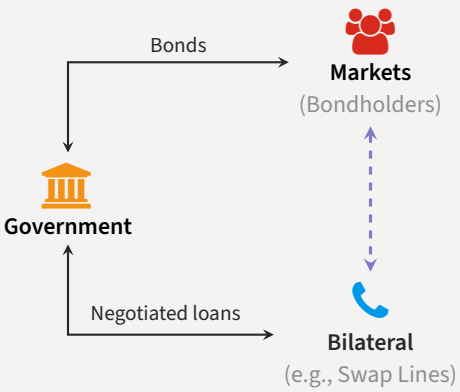


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Relational Overborrowing

Main findings

- Bilateral loans have significant effects on equilibrium outcomes
  - … provide funding when other sources dry up (e.g. because of default risk) ▲
  - … can also incentivize more **risk-taking** ▼

- If the rate on bilateral loans is decreasing in *market* debt [cross-elasticity]
  - … government issues market debt more quickly, delevers more slowly
  - … spends longer in the risky region
  - … defaults more frequently
  - … welfare losses for the government

- Cross-elasticity can emerge endogenously from bargaining
  - … market issuance raises cash-on-hand and strengthens the gov't negotiating stance
  - … **relational overborrowing**

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Literature

- Sovereign debt/default with interactions from ‘official’ debt
  - … senior debt (Hatchondo, Martinez & Önder 2017), senior debt with conditionality (Boz 2011, Fink & Scholl 2016), bailout agencies (Corsetti, Guimarães & Roubini 2006, Kirsch & Rühmkorf 2017, Roch & Uhlig 2018), official debt (Dellas & Niepelt 2016, Arellano & Barreto 2024, Liu, Liu & Yue 2025, Erce, Galli & Yurdagul 2026)

- Data on new official creditors
  - … Horn, Reinhart & Trebesch 2021a, 2021b, Gelpern et al. 2021, Horn, Parks, Reinhart & Trebesch 2023

- Central Bank swap lines
  - … among advanced economies (Bahaj & Reis 2021, Cesa-Bianchi, Eguren-Martin & Ferrero 2022), data for emerging-market borrowers (Perks, Rao, Shin & Tokuoka 2021)

Roadmap

- Theory
  - Bargaining and the Cross-Elasticity
  - The Cross-Elasticity and Debt Dilution
- Quantitative Model
  - Endogenous Bargaining
  - Programming the Large Lender
- Concluding remarks

Theory

Theory

Bargaining and the Cross-Elasticity

Bargaining Induces a Cross-Elasticity

- Two periods $t \in \{0, 1\}$, endowments $y_0 < y_1$
- Government discount factor β, lender's discount factor $\beta_L > \beta$
- **Bargaining** transfer x at $t = 0$ for repayment $x(1 + r)$ at $t = 1$, weight θ for lender

$$\max_{x,r} \mathcal{L}(x, r)^{\theta} \times \mathcal{B}(x, r; y_0, y_1)^{1-\theta}$$

where

$$\mathcal{L}(x, r) = \underbrace{-x + \beta_L x(1 + r)}_{\text{agreement}}$$

$$\mathcal{B}(x, r; y_0, y_1) = \underbrace{u(y_0 + x) + \beta u(y_1 - x(1 + r))}_{\text{agreement}} - \underbrace{(u(y_0) + \beta u(y_1))}_{\text{threat point}}$$

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Bargaining Induces a Cross-Elasticity

- First-order conditions for bargaining problem

$$u'(y_0 + x) = \frac{\theta}{1-\theta} \frac{\mathcal{B}(x, r; y_0, y_1)}{\mathcal{L}(x, r)}$$

Surplus sharing

$$u'(\underbrace{y_0 + x}_{c_0}) = \frac{\beta}{\beta_L} u'\big(\underbrace{y_1 - x(1+r)}_{c_1}\big)$$

Euler

- Can prove that

$$\frac{\partial x}{\partial y_0} \leq 0 \qquad \text{and} \qquad \frac{\partial r}{\partial y_0} \leq 0$$

- **Later** y_0, y_1 are given by market debt dynamics, so
  - Recent issuances: increases y_0 and decreases y_1, makes you stronger
  - Legacy debt: lowers y_0 and y_1, typically makes you weaker

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Theory

The Cross-Elasticity and Debt Dilution

Debt Dilution with Undefaultable Loans

Issue long bonds b at $t = 0$, short bonds d and loans m at $t = 1$, repay at $t = 2$

$$c_0 = q(b+d)b \qquad c_1 = q(b+d)d + \varphi(b,d,m)m \qquad c_2(z) = \begin{cases} y(z) - m - (b+d), & \text{repayment} \\ y(z) - m - \xi(z), & \text{default} \end{cases}$$

with

$$q(b+d) = \mathbb{P}\left(b + d \leq \xi(z)\right) \qquad \text{and} \qquad m \leq \bar{m}$$

to maximize $\sum_{t=0}^{2} \mathbb{E}\left[u(c_t)\right]$

Commitment: Choose d internalizing effect on initial prices

$$u'(c_0)\underbrace{q'(b+d)b}_{\text{past prices}} + u'(c_1)\Big(\underbrace{q(b+d)}_{\text{revenue}} + \underbrace{q'(b+d)d}_{\text{current prices}} + \underbrace{\varphi'_d(b,d,m)m}_{\text{cross-elasticity}}\Big) = q(b+d)\mathbb{E}\left[u'(c_2^R)\right]$$

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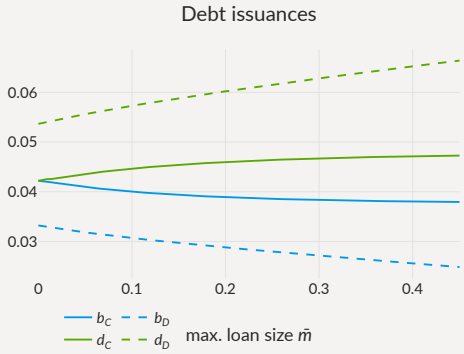
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not under discretion

Debt Dilution with Undefaultable Loans and Cross-Elasticity





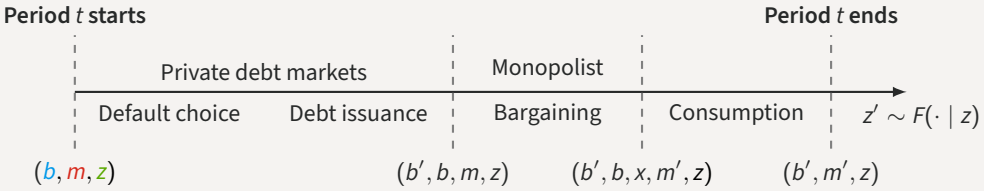
Takeaways

- Bargaining can induce a cross-elasticity from market debt to loan terms
  - … through threat-point manipulation
- The cross-elasticity worsens the debt dilution problem
  - … it can also create welfare losses if strong enough

Quantitative Model

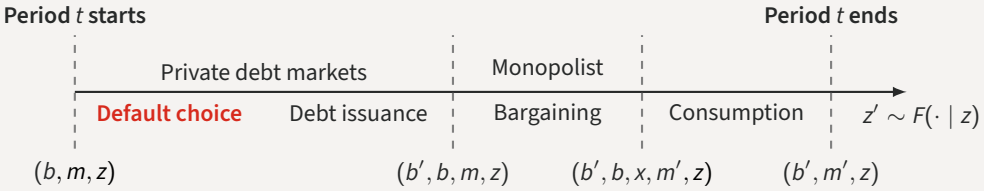
Timeline of Events

- Enter period t owing b to bondholders, m to monopolist, income $y(z)$



Timeline of Events

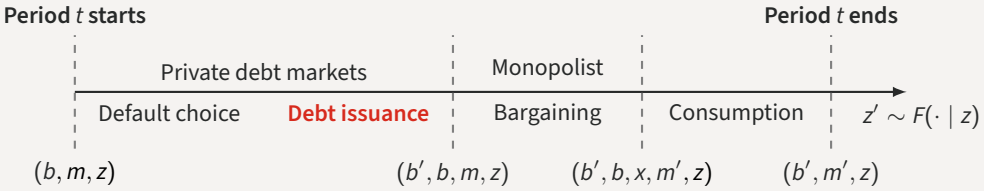
- Choose to repay or default the *market* debt subject to convex output costs



$$v(b, m, z) = \max \left\{ v_R(b, m, z) + \epsilon_R, v_D(m, z) + \epsilon_D \right\}$$

Timeline of Events

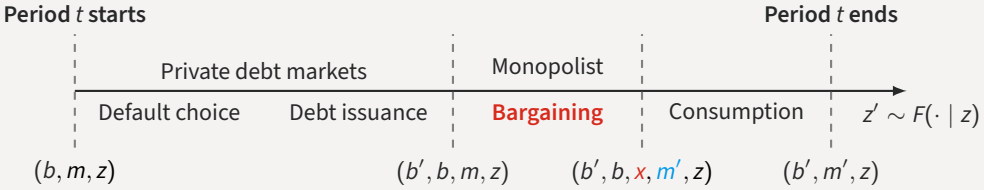
- If repaid, issue new (long-term) debt b' in markets at price q [coupon rate κ, maturity $1/\delta$]



$$q(b', b, m, z) = \beta_L \mathbb{E}\left[(1 - 1_{\mathcal{D}}(b', m', z'))\,(\kappa + (1-\delta) q(b'', b', m', z')) \mid z\right]$$

Timeline of Events

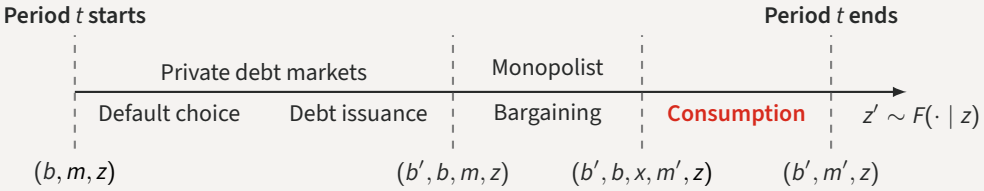
- Meet with senior lender, decide any transfers x and new/remaining balance m'



$$r(x, m', m) = \frac{m'}{x + m} - 1$$

Timeline of Events

- Consume output plus revenues from debt issuance plus transfers minus debt service



$$c_R = y(z) + q(b', b, m, z)(b' - (1 - \delta)b) + x_R(b', b, m, z) - \kappa b$$
$$c_D = y(z) - h(z) + x_D(m, z); \qquad x_D(m, z) \leq \Gamma(m)$$

Calibration

- Standard strategy in sovereign debt: match Argentina 1993-2001 to model w/o swaps

	Parameter	Value
Sov's discount factor	β	0.9607
Sov's risk aversion	γ	2
Pref. scale	χ	0.025
Lender's barg. power	θ	0.5
Risk-free rate	r	0.01
Duration of debt	ρ	0.05
Income autocorr.	ρ_z	0.9484
Std. dev. y_t	σ_z	0.02
Reentry prob.	ψ	0.0385
Default cost: linear	d_0	-0.2514
Default cost: quadratic	d_1	0.292

	Data	Only market
Avg spread (bps)	815	740
Std spread (bps)	443	485
Debt-to-GDP (%)	17.4	18.1

Note moments from pre-default samples

Quantitative Model

Endogenous Bargaining

Bargaining Stage with Senior Lender

- At state z, owing debt b bonds and m on the loan and having issued b'

$$\max_{x,m'} \mathcal{L}_R(b', x, m, m', z)^{\theta} \times \mathcal{B}_R(b', b, x, m, m', z)^{1-\theta}$$

Lender surplus

Government surplus

- Lender's surplus

$$\mathcal{L}_R(b', x, m, m', z) = \underbrace{\left(-x + \beta_L \mathbb{E}\left[h(b', m', z') \mid z\right]\right)}_{\text{agreement}} - \underbrace{\left(m + \beta_L \mathbb{E}\left[h(b', 0, z') \mid z\right]\right)}_{\text{threat point}}$$

- Government's surplus

$$\mathcal{B}_R(b', b, x, m, m', z) = \underbrace{u\left(y(z) + B(b', b, m, z) + x\right) + \beta \mathbb{E}\left[v(b', m', z') \mid z\right]}_{\text{agreement}}$$

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same sdf as markets

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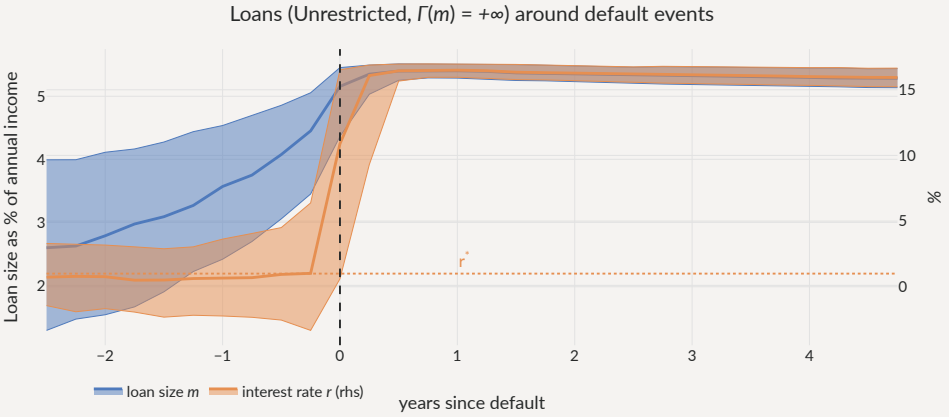
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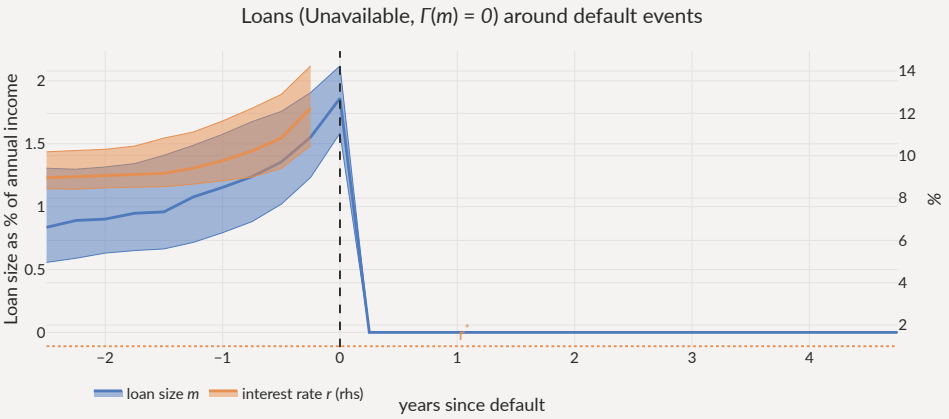
Usage of Loans

- Sharp acceleration before defaults, maxed out *in* default if possible



Usage of Loans

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Limiting Loans in Default

- **Limited**: cannot borrow in default $\Gamma(m) = m$ **Unavailable**: $\Gamma(m) = 0$

	Only market	**Unrestricted,** $\Gamma(m) = +\infty$	**Limited,** $\Gamma(m) = m$	**Unavailable,** $\Gamma(m) = 0$
Avg spread (bps)	630	1,398	1,209	650
Std spread (bps)	474	894	860	381
$\sigma(c)/\sigma(y)$ (%)	104	99	100	104
Debt to GDP (%)	16.4	17.6	17.6	18
Loan to GDP (%)	0	2.44	2.22	0.903
Loan spread (bps)	–	-298	-126	523
Corr. loan & spreads (%)	–	68.9	67.7	63.8
Default frequency (%)	5.03	9.55	9.04	5.28
Welfare gains (rep)	–	-0.095%	-0.038%	0.12%

Note Statistics from each ergodic distribution

Limiting Loans in Default

- Default rates: ↑ with loans (less with Limited/Unavailable)

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Note Statistics from each ergodic distribution

Limiting Loans in Default

- Spreads: ↑ with loans (less with Limited/Unavailable)

	Only market	Unrestricted, $\Gamma(m) = +\infty$	Limited, $\Gamma(m) = m$	Unavailable, $\Gamma(m) = 0$
Avg spread (bps)	630	1,398	1,209	650
Std spread (bps)	474	894	860	381
$\sigma(c)/\sigma(y)$ (%)	104	99	100	104
Debt to GDP (%)	16.4	17.6	17.6	18
Loan to GDP (%)	0	2.44	2.22	0.903
Loan spread (bps)	–	-298	-126	523
Corr. loan & spreads (%)	–	68.9	67.7	63.8
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Limiting Loans in Default

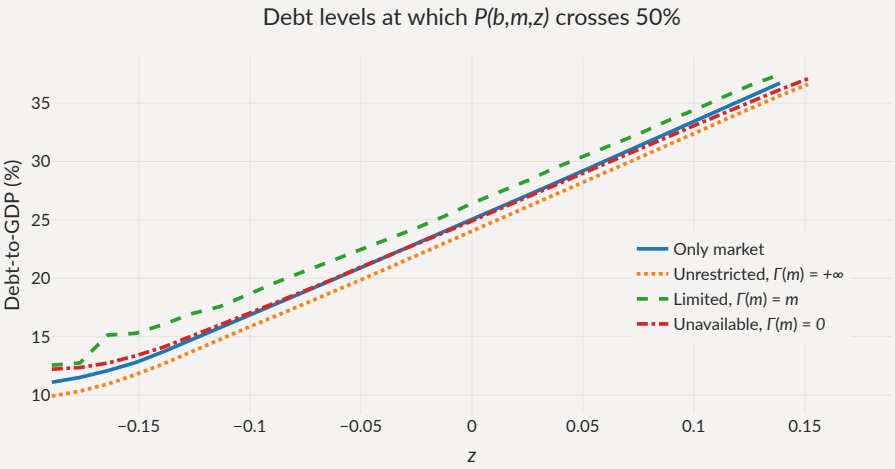
- Welfare: $\approx$ with loans (Unavailable $\succcurlyeq$ Limited $\succcurlyeq$ Unrestricted)

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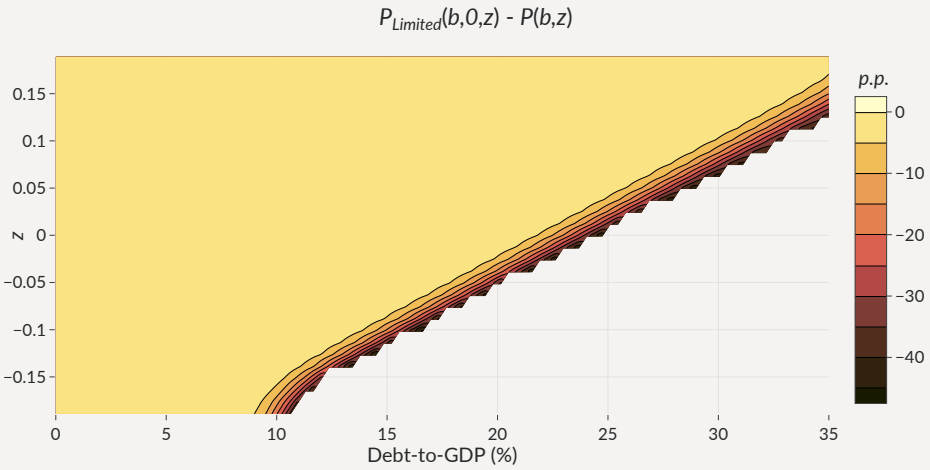
Default Barriers with Loans

- Debt capacity changes marginally; does **not** explain the change in default frequency



Default Probabilities with Loans

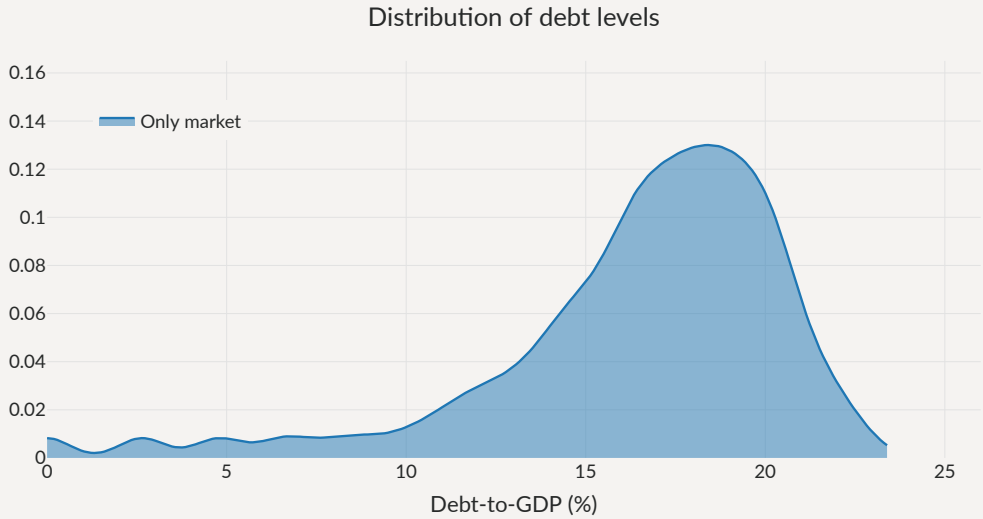
- Loans reduce the default probability everywhere, esp. close to the original barrier



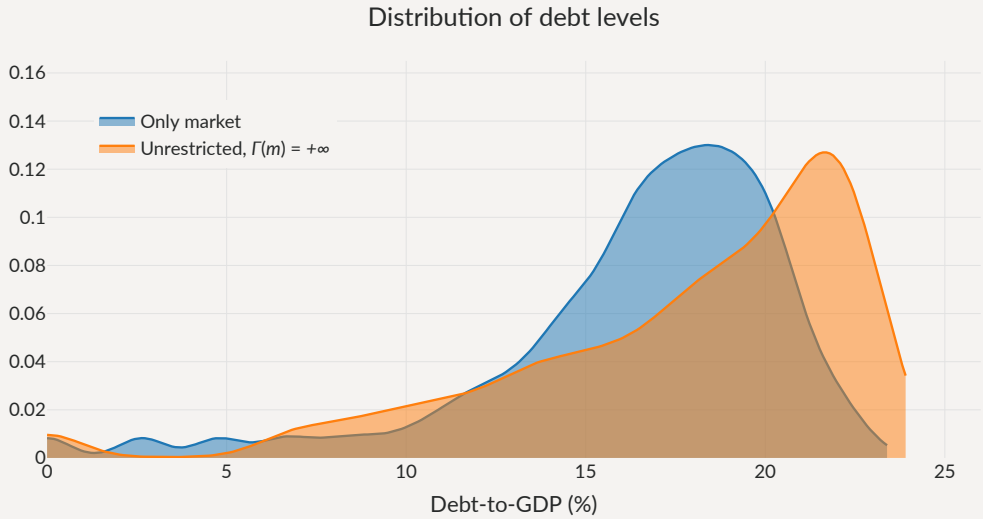
If Loans help repay the debt,

Why are there more defaults with loans?

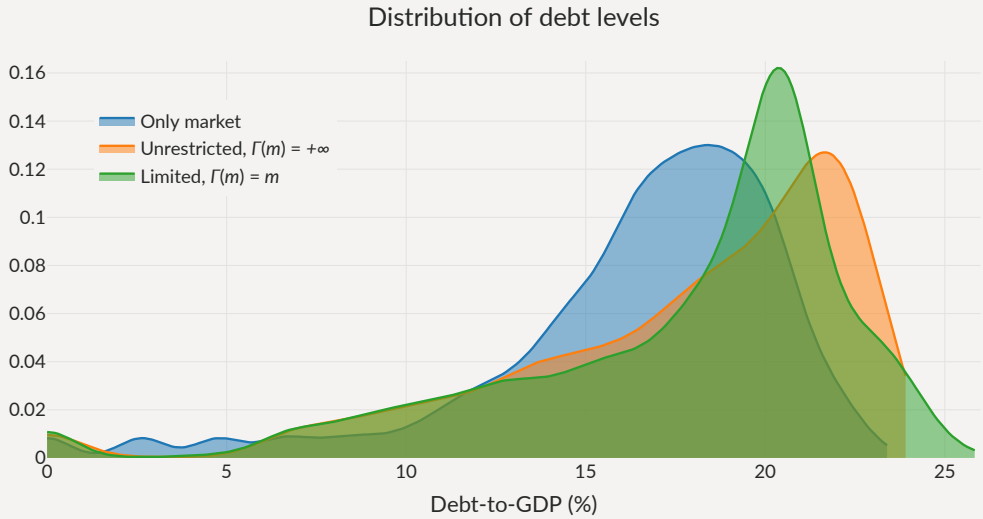
Debt Levels with Loans



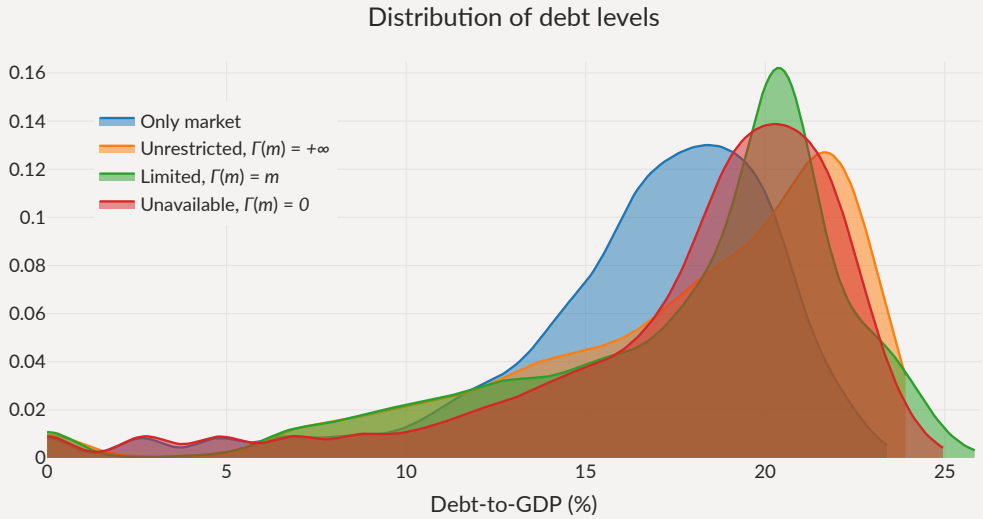
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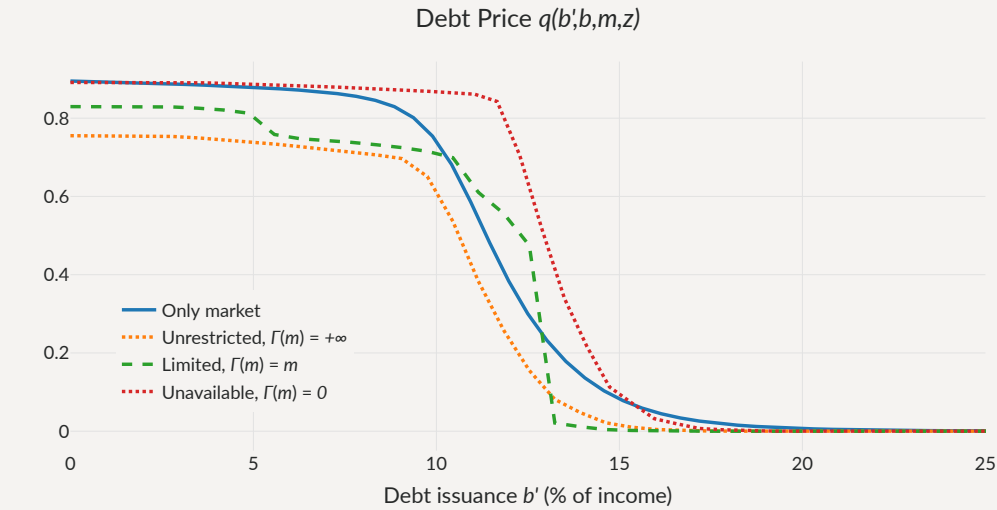


Debt Levels with Loans



Debt Prices with Loans

Lower prices with same ex-post default rates: relational overborrowing similar to debt dilution



Relational Overborrowing

Government's surplus

$$\mathcal{B}_R(b', b, x, m, m', z) = u\big(\overbrace{y(z) + B(b', b, m, z) + x}^{=c_A}\big) + \beta\mathbb{E}\left[v(b', m', z') \mid z\right] - \Big(u\big(\underbrace{y(z) + B(b', b, m, z) - m}_{=c_T}\big) + \beta\mathbb{E}\left[v(b', 0, z') \mid z\right]\Big)$$

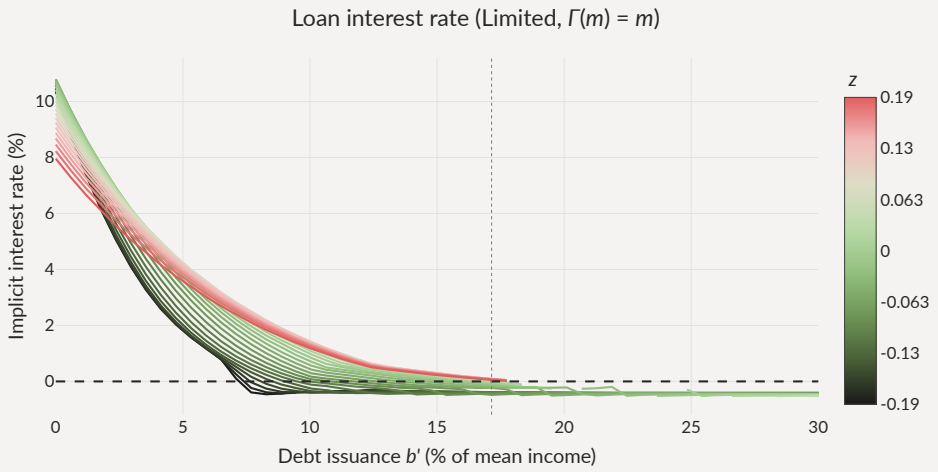
$$u\left(c_A\right) - u\left(c_T\right) \simeq u'(y(z) + B(b', b, m, z) - m) \cdot (x + m)$$

- Revenues from debt issuance $B(b', b, m, z)$ modulate the value of the threat point
  - After large revenues (high q, high b'), gov't flush with cash, **strong** in bargaining
  - After bad issuance (low q *or* low b'), gov't **weak** in bargaining

Risk-taking Incentives

Drawings

Threat point manipulation: increase market b' to reduce bilateral r **Cross-elasticity**



Government's Euler equation for bonds

$$u'(c)\left(q + \frac{\partial q}{\partial b'} i + \frac{1}{1+r}\frac{\partial m'}{\partial b'} + \frac{\partial \frac{1}{1+r_b}}{\partial b'} m'\right) = \beta \mathbb{E}\left[v_b(b', m', z') + v_m(b', m', z')\frac{\partial m'}{\partial b'} \mid z\right]$$

- An extra bond:
  - … yields marginal revenue $q + \frac{\partial q}{\partial b'} i$ [standard debt-Laffer curve effect]
  - … makes you **stronger** in bargaining now
    - … affects the amount m' borrowed from senior lender
    - … affects terms of borrowing with senior lender
  - … makes you **weaker** in bargaining later

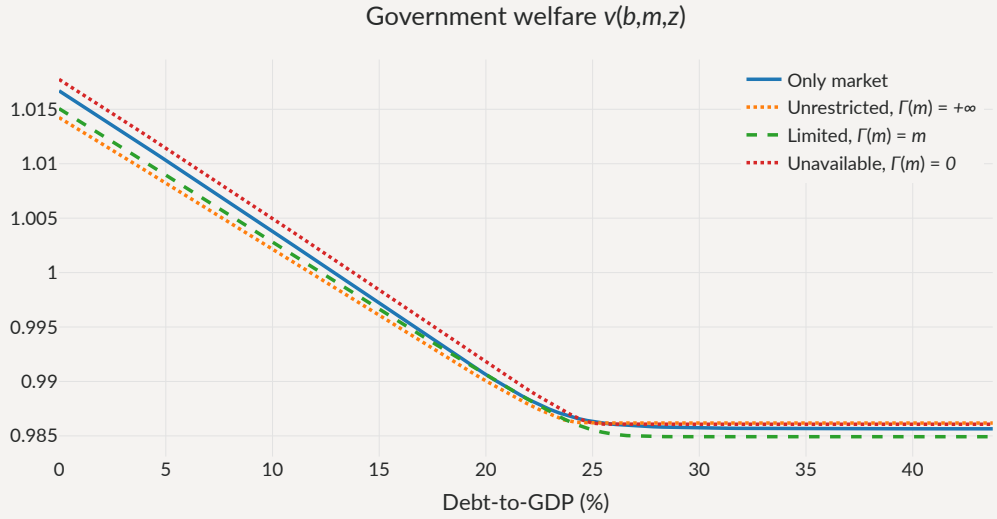
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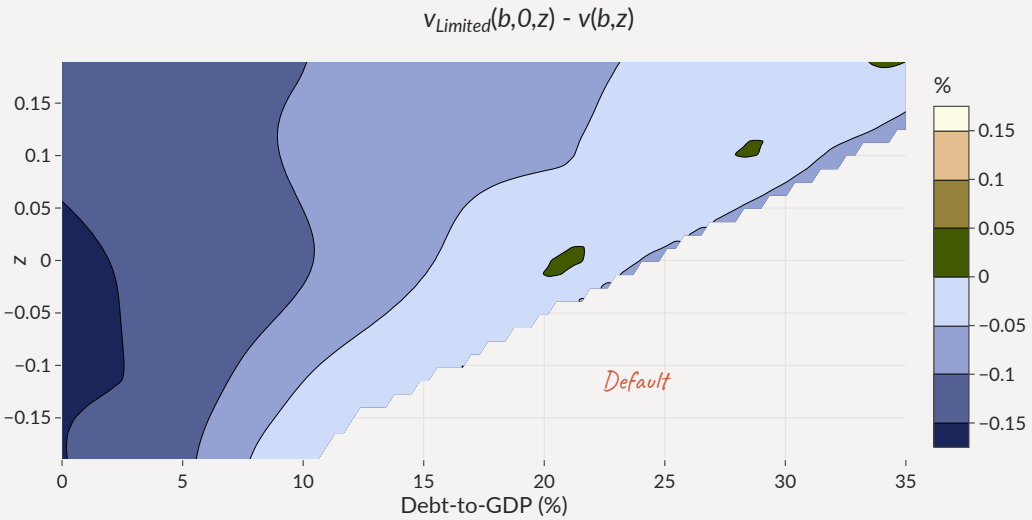
Welfare Effects of Bilateral Loans

Limited $\succcurlyeq$ Unrestricted, but…



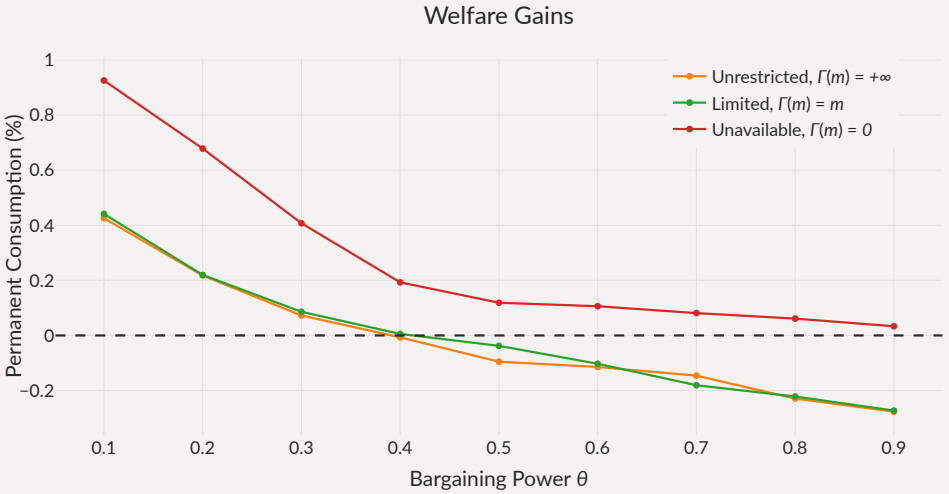
Welfare Effects of Bilateral Loans (cont'd)

Welfare gains overall near barrier: postponing default dominates



Bargaining Power and Welfare

Welfare gains dissipate quickly as the senior lender's bargaining weight rises



Quantitative Model

Programming the Large Lender

Possible Rules

- Explore interest rate rules of the form

$$r(b', m') = \max\{r^\star, \alpha_0 + \alpha_b b' + \alpha_m m'\}$$

- Three versions

Size-dependent
$\alpha_0 > 0, \alpha_b = 0, \alpha_m > 0$
(no cross-elasticity)

Risk-inducing
$\alpha_0 > 0, \alpha_b < 0, \alpha_m \geq 0$
(mimics eq'm w. bargaining)

Optimal
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Equilibrium with Exogenous Rules

- 'Only market' standard calibration to Argentina 1993-2001

	Only market	Size dependent r	Risk inducing r	Optimal r
Avg spread (bps)	630	413	831	227
Std spread (bps)	474	240	447	130
$\sigma(c)/\sigma(y)$ (%)	104	105	104	106
Debt to GDP (%)	16.4	17.3	17.1	17.7
Loan to GDP (%)	0	0.727	0.486	2.56
Loan spread (bps)	–	662	296	516
Corr. loan & spreads (%)	–	49.4	-29.2	38.9
Default frequency (%)	5.03	3.2	6.19	1.92
Welfare gains (rep)	–	0.15%	-0.097%	0.69%

Note Statistics from each ergodic distribution

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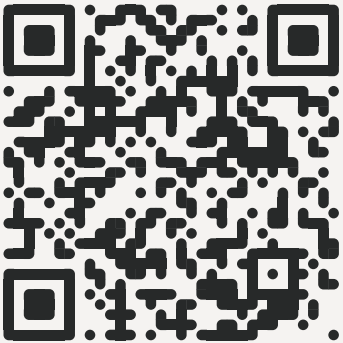
Concluding remarks

The Perils of Bilateral Sovereign Debt

- Simple model of borrowing from markets and a senior bilateral lender
  - … strong interaction between two markets for sovereign debt
  - … even if bilateral loans are **not** used intensely on the equilibrium path

- **Dangerous** when bilateral interest rate responds negatively to *market* debt
  - … the cross-elasticity erodes the disciplining role of spreads
  - … sequential bargaining is a plausible source of the cross-elasticity

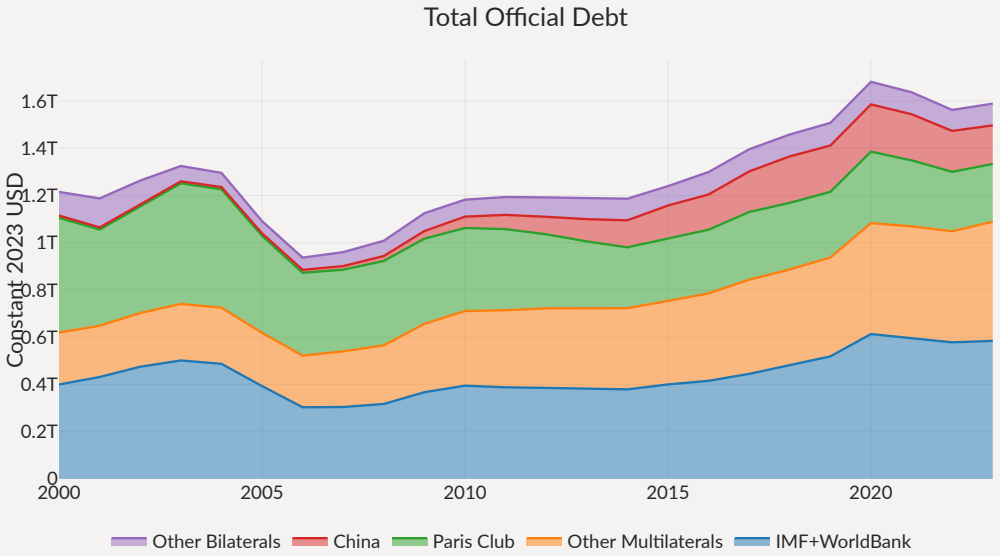
- Transparent, pre-specified **rules** can reverse the force
  - … rules with opposite cross-elasticity improve welfare without losses for lender
  - … **Practical diagnostic:** ask how bilateral rates respond to market issuance or spreads



Scan to find the paper

A New Landscape for Official Sovereign Debt

Back



Risk-taking Incentives

Back

